

Ahmed Hamdy

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SUMMARY

Final-year Software Engineering student at UTM (CGPA: 3.74, Dean's List) with 6 months of production engineering experience at PETRONAS Digital shipping CI/CD pipelines, Terraform-provisioned infrastructure, and cloud-native logging for a 5-service oil and gas platform. Built and deployed a production-grade 9-service microservices platform end-to-end: Stripe payments, RabbitMQ messaging, Redis authentication, CNN inference pipeline, and a hub-and-spoke GitHub Actions CI/CD system across 9 repositories. Currently migrating the platform to Kubernetes with Helm and ArgoCD. Targeting Software Engineering or DevOps/Platform Engineering roles where backend depth and infrastructure ownership intersect.

EDUCATION

Universiti Teknologi Malaysia (UTM)

Johor Bahru, Malaysia

Bachelor of Computer Science (Software Engineering) — CGPA: 3.75/4.00

Expected Oct. 2026

- **Dean's List High Honors**
- Relevant Coursework: Data Structures & Algorithms, Software Engineering, Quality Assurance, Distributed Systems
- Recognition: Exhibited StyleSense at **DIGITEX ASCEND 2030** national technology exhibition

PROFESSIONAL EXPERIENCE

Software Engineering & DevOps Intern

Sep. 2025 – Mar. 2026

PETRONAS Digital

Kuala Lumpur, Malaysia

- Designed and deployed end-to-end CI/CD pipelines in **Azure DevOps** automating **Hasura GraphQL** metadata migration across Development and Production environments, eliminating 100% of manual migration errors across a 5-service oil and gas microservices platform
- Provisioned cloud infrastructure as code using **Terraform** and Azure DevOps YAML pipelines, authoring reusable modules for Linux VM agent pools across restricted enterprise environments
- Engineered a cloud-native centralized logging system using **Serilog** integrated with Azure services and **AWS CloudWatch**, implementing circuit breaker patterns and config-driven provider switching for production observability
- Built a reusable API test automation framework in **Groovy/Spock** covering schema validation, nested JSON normalization, and Dev-vs-Prod regression comparison, integrated into Azure DevOps CI pipelines across 5 microservices
- Authored full-stack test suites for **.NET Core** backend services and **Angular** frontend components using dependency injection, repository mocking, and DOM event testing; delivered across 2-week Agile sprints using Azure Boards

PROJECTS

StyleSense – AI Salon & Barber Booking Platform

Jan. 2025 – Present

- *React, Node.js, FastAPI, PostgreSQL, Docker, RabbitMQ, Redis, Stripe, GitHub Actions, DigitalOcean*
- Architected and deployed a production-grade 9-service microservices platform across 14 **Docker** containers on **DigitalOcean**, serving end-to-end salon booking with Stripe payments, real-time SSE notifications, and AI-driven recommendations
- Implemented **Stripe Connect** two-phase payment flow with manual capture, idempotent webhook processing, and a booking state machine enforcing transactional consistency across distributed service boundaries
- Designed **RabbitMQ** topic-exchange pub/sub event bus and **Redis** phantom-token authentication layer across 8 backend services; built hub-and-spoke **GitHub Actions** CI/CD architecture spanning 9 repositories with an orchestrator automating cross-service deployments
- Trained an **Xception** CNN face-shape classifier on 4,000 images using **TensorFlow/Keras** and **MediaPipe** Face Mesh, integrated with **Groq LLM** API for personalized style recommendations at inference time

Kubernetes Platform Migration – StyleSense

Aug. 2026 – Present

- *Kubernetes, Helm, ArgoCD, Sealed Secrets, k3d, k3s, GitOps*
- Migrating StyleSense's 9-service Docker Compose deployment to production Kubernetes: **Helm** charts authored for all 9 services, **ArgoCD** GitOps wiring in progress, secrets management via **Sealed Secrets**, targeting k3s VPS deployment with full observability stack (Prometheus + Grafana)

Real-Time Analytics System

Jun. 2024 – Aug. 2024

- *Apache Kafka, Zookeeper, Flask, Java, Maven, Docker*
- Engineered a distributed event streaming system using **Apache Kafka** and Zookeeper processing 1,000+ metrics/second with fault-tolerant message delivery and Docker-orchestrated multi-service deployment
- Built a **Flask** REST API serving real-time aggregated data to a live JavaScript dashboard; implemented health-check automation and service recovery logic to minimize downtime

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, Java, C#, Groovy, SQL, Bash

Frameworks: Node.js, Express.js, React, FastAPI, Flask, Spring Boot, .NET Core

DevOps & Cloud: Docker, Kubernetes, Helm, ArgoCD, GitHub Actions, Azure DevOps, Terraform, Azure (Blob Storage, Pipelines, Container Registry), AWS CloudWatch, Linux, Git, GitOps, CI/CD, IaC

Databases: PostgreSQL, MongoDB, Redis, Firebase, Hasura GraphQL

Messaging & Streaming: RabbitMQ, Apache Kafka, Zookeeper

Testing & QA: Spock Framework, Jest, TDD, BDD, Unit/Integration/E2E Testing, API Contract Testing, Mocking, Code Coverage Analysis

ML & AI: TensorFlow, Keras, MediaPipe, PyTorch, OpenCV, Groq API

Methodologies: Microservices, Event-Driven Architecture, REST API Design, Agile/Scrum, SDLC, System Design, Distributed Systems

Also familiar with: Vue.js, Flutter, Dart, AngularJS, C++, Power BI